MILITARY SPECIFICATION

SIGNAL, ILLUMINATION, GROUND, XM199 RED, XM201 WHITE, XM200 GREEN AND XM198 AMBER PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts and the loading, assembling and packing of four signals of different colors designated as Signal, Illumination, Grd, XM199 Red, XM201 White, XM200 Green, and XM198 Amber.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-P-223 - Powder, Black.
MIL-P-2550 - Ammunition and Special Weapons, General Specification for.
MIL-P-20444- Primer, Percussion, M42.

STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1370

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MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-171 - Finishing of Metal and Wood Surfaces
MIL-STD-286 - Propellants, Solid: Sampling, Inspection and Testing

MIL-STD-331 - Fuze and Fuze Components; Environmental and Performance Tests for.
MIL-STD-1167 - Ammunition Data Cards.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1234 - Pyrotechnics; Sampling, Inspection and Testing.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

9201268 - Procedure for Light Output Measurements.
9201392 - Procedure for Color Value Measurement.
9248243 - Signal, Illumination, Grd, Red, XM199
9248316 - Signal, Illumination, Grd, Green, XM200
9248317 - Signal, Illumination, Grd, White, XM201
9248318 - Signal, Illumination, Grd, Amber, XM198
9280090 - Packing and Marking for Box, Wood Wirebound Ammunition for Signal, Illumination Ground
9280092 - Packing and Marking for Shipping and Storage Container
9280095 - Packing and Marking for Sleeve.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The assembly shall comply with all the requirements specified on Drawings (dwgs.) 9248243, 9248316, 9248317, 9248318 and with all requirements specified in applicable specifications.

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3.3 Moisture content

3.3.1 Moisture content of signal composition.-The moisture content of each illuminant composition at the loading station at the time of loading shall not exceed 0.1 percent.

3.3.2 Moisture content of black powder.-The moisture content of the black powder at the loading station at the time of loading shall not exceed 0.3 percent.

3.4 Color value

3.4.1 Red Illuminant Assembly.-The color value of each red illuminant assembly shall be not less than .600.

3.4.2 Green Illuminant Assembly.-The color value of each green illuminant assembly shall be not less than .400.

3.4.3 Amber Illuminant Assembly.-The color value of each amber illuminant assembly shall be not less than .050.

3.5 Candlepower

3.5.1 Red, Green, Amber and White Illuminant Assembly.-The candlepower of each red, green, amber and white illuminant assembly shall be not less than 5000 candlepower.

3.6 Functioning.-The signal assembly shall ignite and shall comply with the following requirements.

3.6.1 Premature.-The signal assembly shall not burst in the projector and shall not fail to reach a minimum height of fifteen (15) feet from the projector.

3.6.2 Burning time.-The illuminant assembly shall burn for not less than six (6) seconds and shall burn out prior to hitting the ground.

3.6.3 Altitude.-The illuminant assembly shall produce the correct color trace to a minimum altitude of three hundred (300) feet.

3.7 Water immersion.-After the signals have been immersed in water, the signals shall comply with the requirements of 3.6.1, 3.6.2, and 3.6.3.

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3.8 Cold temperature (minus 65 plus or minus 5 degrees Fahrenheit (F.).-The signal assemblies shall comply with the requirements of 3.6.1, 3.6.2, and 3.6.3.

3.9 Hot temperature (160 plus or minus 5 degrees Fahrenheit (F.).-The signal assemblies shall comply with the requirements of 3.6.1, 3.6.2, and 3.6.3.

3.10 Temperature and humidity.-After the signals have been subjected to the temperature and humidity conditioning, the signal assemblies shall comply with the requirements of 3.6.1, 3.6.2, and 3.6.3.

3.11 Transportation-vibration.-The signals shall withstand transportation-vibration without exploding or burning. After the signals have been subjected to transportation-vibration, the signal assemblies shall comply with the requirements of 3.6.1, 3.6.2, and 3.6.3.

3.12 Workmanship.-All parts shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of chips, burrs, sharp edges, cracks, dirt, grease, rust and other foreign material. The metal parts shall not contain metal defects such as inclusions, cold shuts, or porous areas. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.13 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the

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supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents and that certificate of conformance and statements of findings on all compositions and powders are on file and available for review. Statements of findings on moisture content for all materials shall be submitted with each lot.

d. Number of items in the lot

e. Date submitted

f. A statement that the components and assemblies were dried and assembled as required by the drawing.

The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-Prior to the start of regular production, the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the Contracting

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Officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of inert parts, sub-assemblies and signal assemblies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government. Unless otherwise directed by the contracting officer, loaded components shall not be submitted for any test until the inert components first article has been accepted.

4.2.2 First article inspection provisions

4.2.2.1 Inert parts.-Twenty-five (25) complete sets of inert parts (i.e. 25 each of every inert component and inert subassembly) shall be submitted to a Government approved laboratory designated by the Contracting Officer. The inert parts will be subjected to any or all of the examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all of the requirements of the applicable drawings. If any inert part fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an inert component or subassembly to comply with any of the stated requirements.

4.2.2 Illuminant assembly.-Twenty-five (25) assemblies of each color shall be submitted to a Government approved laboratory for tests in accordance with the method specified in

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4.4.2 and 4.4.3. If any assembly fails to comply with the specific requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an illuminant assembly to comply with any of the stated requirements.

4.2.2.3 Loaded assemblies.-Fifty (50) loaded assemblies of each color completely packaged and sealed as required by the applicable drawings shall be submitted to a Government proving ground designated by the Contracting Officer. The assemblies shall be subjected to the tests specified in 4.4.4, 4.4.5, and 4.4.6. The quantities for test shall be as follows:

a. Ten (10) at water immersion (see 3.7).
b. Ten (10) at cold temperature (see 3.8).
c. Ten (10) at hot temperature (see 3.9).
d. Ten (10) for temperature humidity (see 3.10)
e. Ten (10) for transportation-vibration (see 3.11)

The first article sample shall be rejected if a critical defect occurs or if any major defect as classified in Table III occurs. The Government reserves the right to terminate its inspection upon encountering any failure of a signal to comply with any of the stated requirements.

4.3 Inspection provisions.

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications,

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or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of signals assemblies shallccontain:

a. Parts of one interfix lot number from one supplier.
b. Primers of one interfix lot number from one manufacturer.
c. Signals of one color only.
d. Composition ingredients from not more than one lot from one supplier.
f. Illuminant composition produced by one manufacturer under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cap (see dwg. 9248314 covering a detail of dwgs. 9248243, 9248316, 9248317 and 9248318).

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Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Crack or split	Visual	01001
Minor:	AQL 1.50 percent		
201.	Outside diameter of case contact surface	Gage	01002
202.	Protective coating missing or incomplete	Visual	01003
203.	Overall depth	Gage	01004
204.	Evidence of poor workmanship (except cracks as defined above) (see 3.12)	Visual	01005

4.3.2.2 Cup (see dwg. 9248251 covering a detail of dwgs. 9248243, 9248316, 9248317 and 9248318).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Crack or split	Visual	02001
Major: None defined.			
Minor:	AQL 2.50 percent		
201.	Total length	Gage	02002
202.	Outside diameter	Gage	02003
202.	Outside diameter	Gage	02003
203.	Thickness through bottom	Gage	02004
204.	Wall thickness	Gage	02005
205.	Evidence of poor workmanship (except cracks as defined above) (see 3.12)	Visual	02006

4.3.2.3 Case, signal, red (see dwg. 9248310 covering a detail of dwg. 9248243) when applicable.

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Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Crack or split	Visual	03001
2.	Color identification incorrect or knurl improper	Visual	03002
Major:	AQL 1.00 percent		
101.	Flash hole obstructed	Visual	03003
102.	Pitch diameter of thread (max)	Gage	03004
103.	Major diameter of thread (max)	Gage	03005
104.	Diameter of charge cavity	Gage	03006
105.	Concentricity of inside diameter with outside diameter	Gage	03007
106.	Concentricity of primer cavity with thread	Gage	03008
Minor:	AQL 4.00 percent		
201.	Total length	Gage	03009
202.	Depth of primer cavity	Gage	03010
203.	Depth of charge cavity	Gage	03011
204.	Large outside diameter (after knurling)	Gage	03012
205.	Diameter of primer cavity	Gage	03013
206.	Outside diameter at primer end	Gage	03014
207.	Length to threaded end	Gage	03015
208.	Major diameter of thread, min	Gage	03016
209.	Pitch diameter of thread, min	Gage	03017
210.	Bare spot in protective coating or coating missing	Visual	03018
211.	Evidence of poor workmanship (except cracks as defined above) (see 3.12)	Visual	03019

4.3.2.4 Case, signal, green (see dwg. 9248311 covering a detail of dwg. 9248316) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Crack or split	Visual	04001
2.	Color incorrect or knurled, bands improper	Visual	04002

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Major:	AQL 1.00 percent		
101.	Flash hole obstructed	Visual	04003
102.	Pitch diameter of thread (max)	Gage	04004
103.	Major diameter of thread (max)	Gage	04005
104.	Diameter of charge cavity	Gage	04006
105.	Concentricity of inside diameter with outside diameter	Gage	04007
106.	Concentricity of primer cavity with thread	Gage	04008
Minor:	**AQL 4.00 percent**		
201.	Total length	Gage	04009
202.	Depth of primer cavity	Gage	04010
203.	Depth of charge cavity	Gage	04011
204.	Large outside diameter (after knurling)	Gage	04012
205.	Diameter of primer cavity	Gage	04013
206.	Outside diameter at primer end	Gage	04014
207.	Length to threaded end	Gage	04015
208.	Major diameter of thread, min	Gage	04016
209.	Pitch diameter of thread, min	Gage	04017
210.	Bare spot in protective coating or coating missing	Visual	04018
211.	Evidence of poor workmanship (except cracks as defined above) (see 3.12)	Visual	04019

4.3.2.5 Case, signal, white (see dwg. 9248312 covering a detail of dwg. 9248317) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Crack or split	Visual	05001
2.	Color incorrect or knurled bands improper	Visual	05002
Major:	**AQL 1.00 percent**		
101.	Flash hole obstructed	Visual	05003
102.	Pitch diameter of thread (max)	Gage	05004
103.	Major diameter of thread (max)	Gage	05005

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104.	Diameter of charge cavity	Gage	05006
105.	Concentricity of inside diameter with outside diameter	Gage	05007
106.	Concentricity of primer cavity with thread	Gage	05008
Minor:	AQL 4.00 percent		
201.	Total length	Gage	05009
202.	Depth of primer cavity	Gage	05010
203.	Depth of charge cavity	Gage	05011
204.	Large outisde diameter (after knurling)	Gage	05012
205.	Diameter of primer cavity	Gage	05013
206.	Outside diameter at primer end	Gage	05014
207.	Length to threaded end	Gage	05015
208.	Major diameter of thread, min	Gage	05016
209.	Pitch diameter of thread, min	Gage	05017
210.	Bare spot in protective coating or coating missing	Visual	05018
211.	Evidence of poor workmanship (except cracks as defined above) (see 3.12)	Visual	05019

4.3.2.6 Case, signal, Amber (see dwg. 9248313 covering a detail of dwg. 9248318) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Crack or split	Visual	06001
2.	Color incorrect or knurled band improper	Visual	06002
Major:	AQL 1.00		
101.	Flash hole obstructed	Visual	06003
102.	Pitch diameter of thread (max)	Gage	06004
103.	Major diameter of thread (max)	Gage	06005
104.	Diameter of charge cavity	Gage	06006
105.	Concentricity of inside diameter with outside diameter	Gage	06007
106.	Concentricity of primer cavity with thread	Gage	06008

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Minor:	AQL 4.00 percent		
201.	Total length	Gage	06009
202.	Depth of primer cavity	Gage	06010
203.	Depth of charge cavity	Gage	06011
204.	Large outside diameter (after knurling)	Gage	06012
205.	Diameter of primer cavity	Gage	06013
206.	Outside diameter at primer end	Gage	06014
207.	Length to threaded end	Gage	06015
208.	Major diameter of thread, min	Gage	06016
209.	Pitch diameter of thread, min	Gage	06017
210.	Bare spot in protective coating or coating missing	Visual	06018
211.	Evidence of poor workmanship (except cracks as defined above) (see 3.12)	Visual	06019

4.3.2.7 Case, assembly, Red (see dwg. 9248253 covering a

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Primer above flush	Visual/ Straight Edge	07001
Major:	AQL 0.40 percent		
101.	Sealing composition not 360° or where not required	Visual	07002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	07003

4.3.2.8 Case, assembly, Green (see dwg. 9248254 covering a detail of dwg. 9248316) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Primer above flush	Visual-Manual Straight edge	08001
Major:	AQL 0.40 percent		
101.	Sealing composition not 360° or where not required	Visual	08002

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	08003

4.3.2.9 Case, assembly, White (see dwg. 9248255 covering a detail of dwg. 9248317) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Primer above flush	Visual-Manual Straight edge	09001
Major:	AQL 0.40 percent		
101.	Sealing composition not 360° or where not required	Visual	09002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	09003

4.3.2.10 Case, assembly, Amber (see dwg. 9248256 covering a detail of dwg. 9243818) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Primer above flush	Visual-Manual Straight edge	10001
Major:	AQL 0.40 percent		
101.	Sealing composition not 360° or where not required	Visual	10002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	10003

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4.3.2.11 Illuminant assembly, prior to loading applicable illuminant composition (see dwg. 9248244 covering a detail of dwgs. 9248243, 9248316, 9248317 and 9248318).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Drop shot missing or inadequate....	Visual-Scale	11001
Minor: None defined.			

4.3.2.12 Illuminant assembly (see dwg. 9248244 covering a detail of dwgs. 9248243, 9248316, 9248317 and 9248318).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Any illuminant composition increment missing (color as applicable).....	Visual	12001
102.	Black powder missing.....	Visual	12002
Minor:	AQL 1.00 percent		
201.	Depth to charge, min.....	Gage	12003
202.	Evidence of poor workmanship (see 3.12).....	Visual	12004

4.3.2.13 Signal, Illumination, Ground, Red, XM199, prior to assembling illuminant assembly, red (see dwg. 9248243).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Propellant charge missing.....	Visual	13001

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Minor: None defined.

4.3.2.14 Signal, Illumination, Ground, Red, XM199, prior to assembling wad and cap (see dwg. 9248243).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color of illuminant assembly and case assembly do not coincide (color and knurling).....	Visual	14001
2.	Illuminant assembly binds in case assembly.....	Visual Manual	14002
Major: None defined.			
Minor: None defined.			

4.3.2.15 Signal, Illumination, Ground, Red, XM199 (see dwg. 9248243).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Any component missing or improperly assembled.....	Visual	15001
Minor:	AQL 1.00 percent		
201.	Marking missing, misleading or unidentifiable.....	Visual	15002
202.	Evidence of poor workmanship (see 3.12).....	Visual	15003

4.3.2.16 Signal, Illumination, Ground, Green, XM200, prior to assembling illuminant assembly, green (see dwg. 9248316).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			

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Major:	AQL 0.40 percent		
101.	Propellant charge missing.....	Visual	16001

Minor: None defined.

4.3.2.17 Signal, Illumination, Ground, Green, XM200, prior to assembling of wad and cap (see dwg. 9248316).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color of illuminant assembly and case assembly do not coincide (color and knurling).....	Visual	17001
2.	Illuminant assembly binds in case assembly.....	Visual-Manual	17002

Major: None defined.

Minor: None defined.

4.3.2.18 Signal, Illumination, Ground, Green, XM200 (see dwg. 9248316).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Any component missing or improperly assembled.....	Visual	18001
Minor:	AQL 1.00 percent		
201.	Marking missing, misleading or unidentifiable.....	Visual	18002
202.	Evidence of poor workmanship (see 3.12).....	Visual	18003

4.3.2.19 Signal, Illumination, Ground, White, XM201, prior to assembling illuminant assembly, white (see dwg. 9248317).

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Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Propellant charge missing.....	Visual	19001
Minor: None defined.			

4.3.2.20 Signal, Illumination, Ground, White, XM201, prior to assembling wad and cap (see dwg. 9248317).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color of illuminant assembly and case assembly do not coincide (color and knurling).....	Visual	20001
2.	Illuminant assembly binds in case assembly.....	Visual-Manual	20002
Major: None defined.			
Minor: None defined.			

4.3.2.21 Signal, Illumination, Ground, White, XM201, (see dwg. 9248317).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Any component missing or improperly assembled	Visual	21001
Minor:	AQL 1.00 percent		
201.	Marking missing, misleading or unidentifiable.....	Visual	21002

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202.	Evidence of poor workmanship, (see 3.12).....	Visual	21003

4.3.2.22 Signal, Illumination, Ground, Amber, XM198, prior to assembling illuminant assembly, amber (see dwg. 9248318).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Propellant charge missing.....	Visual	22001
Minor: None defined.			

4.3.2.23 Signal, Illumination, Ground, Amber, XM198, prior to assembling wad and cap (see dwg. 9248318).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color of illuminant assembly and case assembly do not coincide (color and knurling).....	Visual	23001
2.	Illuminant assembly binds in case assembly.....	Visual-Manual	23002
Major: None defined.			
Minor: None defined.			

4.3.2.24 Signal, Illumination, Ground, amber, XM198 (see dwg. 9248318).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			

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Major:	AQL 0.40 percent		
101.	Any component missing or improperly assembled	Visual	24001
Minor:	AQL 1.00 percent		
201.	Marking missing, misleading or unidentifiable	Visual	24002
202.	Evidence of poor workmanship (see 3.12)	Visual	24003

4.3.2.25 Packing and marking for sleeve, prior to sealing (see dwg. 9280095).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Sleeve damaged	Visual	25001
102.	Number of assemblies in sleeve incorrect	Visual	25002
Minor:	None defined.		

4.3.2.26 Packing and marking for sleeve, sealed (see dwg. 9280095).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color identification missing or incorrect	Visual	26001
Major:	AQL 0.40 percent		
101.	Sleeve damaged to extent that contents are exposed or liable to become exposed	Visual	26002
Minor:	AQL 1.00 percent		
201.	Contents loose	Visual	26003
202.	Marking misleading or unidentifiable (except as defined by critical above)	Visual	26004

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4.3.2.27 Packing and marking for shipping and storage container for signal, illumination ground, prior to closing (see dwg. 9280092).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color identification missing or incorrect	Visual	27001
Major:	AQL 0.40 percent		
101.	Container damaged to extent that contents are exposed or liable to become exposed	Visual	27002
Minor:	AQL 0.65 percent		
201.	Number of sleeves incorrect	Visual	27003

4.3.2.28 Packing and marking for shipping and storage container for signal, illumination, ground, sealed (see dwg. 9280092).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color identification missing or incorrect	Visual	28001
Major: None defined.			
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable (except at defined by critical above)	Visual	28002

4.3.2.29 Packing and marking for box, wirebound ammunition for signal, illumination ground, prior to closing (see dwg. 9280090).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Number of shipping and storage containers incorrect	Visual	29001
102.	Any packing material missing	Visual	29002
103.	Foreign material in box	Visual	29003
Minor:	None defined.		

4.3.2.30 Packing and marking for box, wood, wirebound ammunition for signal, illumination ground, sealed (see dwg. 9280090).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color identification missing or incorrect	Visual	30001
Major:	AQL 0.65 percent		
101.	Box damaged exposing contents	Visual	30002
102.	Wire missing, broken or loose	Visual-Manual	30003
103.	Board broken or loose	Visual-Manual	30004
Minor:	AQL 1.50 percent		
201.	Metallic seal missing, unsealed or improperly positioned	Visual	30005
202.	Marking missing, misleading or unidentifiable (except as defined by critical above)	Visual	30006
203.	Contents loose	Manual	30007

4.3.3 Testing

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4.3.3.1 Moisture content (see Table I) - Major defect.

TABLE I

Materials	Code No.
Illuminant composition (red, green, white and amber) (see 3.3.1)	31001
Black powder (see 3.3.2)	31002
Packing material (see dwg. 9280092)	31003

The contractor shall provide adequate controls to insure that the materials comply with the requirements. For verification, the contractor shall select and subject to test, one sample of each material from each eight (8) hours production. A composite sample shall not be used. If the moisture content of the same exceeds the requirement, that quantity or sub-lot material represented by the sample shall not be used in production. If the quantity of material or sub-lot with excessive moisture has been used in loading and packing, the remaining unloaded and unpacked material shall not be used in production, and the loaded and packing signals shall be rejected. Test shall be performed as specified in 4.4.1.

4.3.3.2 Color value and candlepower of illuminant assembly (see Table II) - Major defect.

TABLE II

Defect	Code No.
Color Value	
a. Red (see 3.4.1)	12005
b. Green (see 3.4.2)	12006
c. Amber (see 3.4.3)	12007
Candlepower less than min (see 3.5)	12008

4.3.3.2.1 Twenty (20) illuminant assemblies of each color, including white, shall be selected from each lot for test. If two (2) or more assemblies fail to comply with any of the requirements listed in Table II, the lot shall be rejected. The test procedure shall be as specified in 4.4.2 and 4.4.3 (destructive test).

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4.3.3.3 Functioning (Proving Ground Test).-The signal assemblies shall be observed for any evidence of failure to comply with the requirements as classified in Table III, when tested as specified in 4.4.4 (destructive test).

TABLE III

Defects	Classification	Code No
Signal bursts in launcher (see 3.6.1)	Critical	32001
Altitude less than fifteen (15) feet (see 3.6.1)	Critical	32002
Incorrect color	Critical	32003
Signal fails to ignite (see 3.6)	Major	32004
Altitude less than three hundred (300) feet (see 3.6.3)	Major	32005
Burning time less than six (6) seconds (see 3.6.2)	Major	32006
Signal hits ground burning (see 3.6.2)	Major	32007

4.3.3.3.1 Water immersion (see 3.7)

4.3.3.3.1.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the applicable requirements specified, one hundred and twenty-five (125) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if six (6) or more assemblies exhibit any of the defects listed in Table III.

4.3.3.3.1.2 After three (3) consecutive lots have complied with the acceptance criteria of 4.3.3.3.1.1, fifty (50) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if three (3) or more assemblies exhibit any of the defects listed in Table III.

4.3.3.3.2 Cold temperature (see 3.8).

4.3.3.3.2.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the applicable requirement specified, fifty (50) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if four (4) or more assemblies exhibit any of the defects listed in Table III.

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4.3.3.3.2.2 After three (3) consecutive lots have complied with the acceptance criteria of 4.3.3.3.2.1, thirty-two (32) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if three (3) or more assemblies exhibit any of the defects listed in Table III.

4.3.3.3.3 Hot temperature (see 3.9).

4.3.3.3.3.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the applicable requirements specified, fifty (50) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if four (4) or more assemblies exhibit any of the defects listed in Table III.

4.3.3.3.3.2 After three (3) consecutive lots have complied with the acceptance criteria of 4.3.3.3.3.1, thirty-two (32) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if three (3) or more assemblies exhibit any of the defects listed in Table III.

4.3.3.4 Temperature and humidity (see 3.10).-Thirty-two (32) signal assemblies of each color shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if two (2) or more assemblies exhibit any of the defects listed in Table III, when tested as specified in 4.4.5 (destructive test).

4.3.3.5 Transportation-vibration (see 3.11).-The signal assemblies shall be observed for any evidence of failure to comply with the requirements as classified in Table V, when tested as specified in 4.4.6 (destructive test).

TABLE IV

Defect	Classification	Code No.
Transportation-vibration safety failure (see 3.11)	Critical	33001
Failure to meet function requirements after test (see 3.6 and 4.3.3.3)	See Table III	

4.3.3.5.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the applicable requirements specified, eighty (80) signal assemblies

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shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if four (4) or more assemblies exhibit any of the defects listed in Table III.

4.3.3.5.2 After three (3) consecutive lots have complied with the acceptance criteria of 4.3.3.5.1, thirty-two (32) signal assemblies shall be selected from each lot for test. The lot shall be rejected if a critical defect occurs or if two (2) or more assemblies exhibit any of the defects listed in Table III.

4.3.3.6 Heat seal of sleeve (see dwg. 9280092) - Major defect - Code No. 28005.-Then (10) sealed packages shall be selected from each lot for test. If any heat seal fails to comply with the requirements, the lot shall be rejected. The test shall be as specified in 4.4.7 (non-destructive test).

4.3.3.7 Check-test for deterioration of Primer, Percussion, M42 (see MIL-P-20444), Code No. 35001.-If the total elapsed time between original acceptance of the M42 Primer lot and the assembly of that lot into the delay element exceeds two years; or if the M42 Primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check-test for deterioration specified in MIL-P-20444. The test shall be performed by the contractor on assemblies selected by the Government inspector, at the facility assembling the delay element (see 6.4).

4.3.4 The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" in 4.3.2, and test method 4.4 paragraphs herein (see 6.3).

4.4 Test methods and procedures

4.4.1 Moisture content

4.4.1.1 Illuminant composition.-The method of determining the percentage of moisture in the illuminant composition shall be in accordance with MIL-STD-1234, Method 102.1, except that the temperature shall be 130 degrees plus or minus 5 degrees Centrigrade in lieu of 100 degrees Centigrade.

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4.4.1.2 Black powder.-The method of determining the percentage of moisture in black powder shall be in accordance with MIL-P-223.

4.4.1.3 Packing material.-The moisture content shall be determined in accordance with MIL-STD-1234, Method 102.1, except that a 10 gram sample shall be used and dried for two (2) hours.

4.4.2 Color value.-The illuminant assembly shall be supported in a vertical position with the surface to be burned facing up. The assembly shall be ignited and the color value determined and recorded in accordance with the method of inspection on dwg. 9201392.

4.4.3 Candlepower.-The candlepower test shall be performed concurrently with the color value test as specified in 4.4.2 and in accordance with the method of inspection on dwg. 9201268.

4.4.4 Functioning.-The following functioning tests shall be performed at a Government Proving Ground.

4.4.4.1 Water immersion.-The signal assemblies shall be immersed in water at a temperature of 70 plus or minus 5 degrees F. to a depth of 10 to 12 inches for a minimum of 2 hours. Upon completion of water immersion, the signal assemblies shall be fired vertically from a fixture. The functioning shall be observed for compliance with the requirements.

4.4.4.2 Cold temperature (minus 65 plus or minus 5 degrees F.).-The signal assemblies shall be conditioned for a minimum of sixteen (16) hours at cold temperature and fired within 5 minutes after removal from the conditioning chamber by the same method specified in 4.4.4.1. The functioning shall be observed for compliance with the requirements.

4.4.4.3 Hot temperature (160 plus or minus 5 degrees F.).-The signal assemblies shall be conditioned for a minimum of sixteen (16) hours at hot temperature and fired within 5 minutes after removal from the conditioning chamber by the same method specified in 4.4.4.1. The functioning shall be observed for compliance with the requirements.

4.4.5 Temperature and humidity.-The signal assemblies shall be subjected to a one 14-day temperature and humidity cycle as specified in MIL-STD-331, test 105. The signals shall be fired by the same method specified in 4.4.4.1. The functioning shall be observed for compliance with the requirements.

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4.4.6 Transportation-vibration.-The test shall be performed as specified in Test 104, Procedure II, of MIL-STD-331, at ambient temperature. The assembly, after test, shall be examined visually. If no degredation affecting safety or operability is apparent, the assembly shall be functioned as specified in 4.4.4.1 and observed for compliance with the requirements.

4.4.7 Heat seal of sleeve.-The test shall be performed in accordance with the procedure specified in MIL-P-116. After test the seal shall be examined for compliance with the requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The signal assemblies shall be preserved and packaged in accordance with dwgs. 9280092 and 9280095.

5.2 Packing

5.2.1 Level A.-The signal assemblies shall be packed in accordance with dwg. 9280090.

5.3 Marking.-Labels, containers and exterior wood boxes shall be marked in accordance with dwgs. 9280090 and 9280092.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are desig-

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nated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

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6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Cost of check test.-The contracting officer should arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The tests shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the primers in the simulators and shall not constitute a basis for rejection against either contractor except where deterioration has occured as a direct result of carelessness in handling, storage, etc. permitted while the primer lots were under the jurisdiction of either contractor.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project Number: 1370-A-003

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